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The Impact of Freedom of Speech on Conspiracy Beliefs

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Received: 11 February 2025 | **Revised:** 15 October 2025 | **Accepted:** 15 October 2025

Keywords: conspiracy beliefs | conspiracy mentality | democracy | freedom of speech | mitigation

ABSTRACT

Conspiracy beliefs are often portrayed as a threat to democracies. However, less is known about the extent to which the state of democracy may affect conspiracy beliefs. Hence, we investigated the impact of the societal level of freedom of speech on conspiracy beliefs. In Study 1, using aggregated nation-level data ($N = 68$ countries), we found that conspiracy beliefs were higher in countries with lower population support for and lower expert-estimated freedom of speech. However, this effect was no longer significant after controlling for two confounders: corruption and electoral democracy. In Study 2a ($N = 190$, preregistered), inducing low (vs. high) freedom of speech increased conspiracy beliefs in a fictional society. We replicated this effect in Study 2b ($N = 90$, preregistered), extending it to generic conspiracy beliefs. In Study 3a ($N = 436$, preregistered), we successfully designed a new induction of freedom of speech in an ecological setting but failed to replicate its effect on conspiracy beliefs. We adjusted this new experimental material in Study 3b ($N = 498$, preregistered) and observed higher conspiracy beliefs under low freedom of speech, compared to the high freedom of speech and control conditions, which did not differ. Hence, the effect stemmed from low freedom of speech increasing conspiracy beliefs, rather than high freedom of speech reducing them. In Studies 2–3, the effect of freedom of speech remained significant when controlling for perceptions of democracy, highlighting the unique nature of this societal factor in shaping conspiracy beliefs. We discuss freedom of speech as a way to mitigate conspiracy beliefs.

1 | Introduction

Conspiracy beliefs are accusations that one or more groups are acting in secret to accomplish a goal that will directly or indirectly harm one or more groups (Bertin et al. 2023). The growing interest for the study of these beliefs in social and political psychology has spawned multiple perspectives (Pilch et al. 2023). Such a diversification is necessary to better understand conspiracy beliefs and help prevent their detrimental consequences (Jolley et al. 2022). In this context, several contributions have recently embraced a macro-level of analysis.

The macro-level approach relies on intercultural comparisons to identify factors that can predict higher levels of conspiracy beliefs at the national level (see Douglas et al. 2023; Hornsey, Bierwiaczonek, et al. 2023; Hornsey and Pearson 2022). For example, populations from countries with more corruption (Alper 2023) or engaged in conflicts (Hebel-Sela et al. 2022) have been shown to hold higher levels of conspiracy beliefs. Economic inequalities are also a robust predictor of conspiracy beliefs (Casara et al. 2022; Jetten et al. 2022). Interestingly, both higher perceived and objective economic inequalities (e.g., GDP per capita) predict increased conspiracy beliefs (Hornsey, Pearson, et al. 2023).

In this contribution, we investigated another potential macro-level factor of conspiracy beliefs, namely, the level of freedom of speech in a given society. According to the 19th article of the Universal Declaration of Human Rights, freedom of speech refers

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to 'freedom to hold opinions without interference and to seek, receive and impart information and ideas through any media and regardless of frontiers' (United Nations General Assembly 1948, Article 19). Freedom of speech has been at the core of modern democracies since the Enlightenment (Doré et al. 2022), as reflected by its centrality in the First Amendment of the US Constitution (Downs and Cowan 2012). The impact of the level of freedom of speech and, more generally, of the respect of democratic principles on conspiracy beliefs has been the focus of scant attention.

In contrast, the reverse causal relationship has attracted considerable attention. Several contributions reveal that exposure to conspiracy theories leads to lower levels of political engagement (e.g., voting in the next election; Jolley and Douglas 2014; Imhoff et al. 2021), adherence to democratic norms (e.g., willingness to accept the results of elections; Albertson and Guiler 2020), and support of democratic principles (e.g., equality between citizens despite their differences; Bertlich and Imhoff 2024). This robust effect holds for conspiracy theories specific to the elections (Albertson and Guiler 2020) but also famous conspiracy theories (i.e., Lady Diana's death; Jolley and Douglas 2014) and generic conspiracy theories (i.e., conspiracy mentality; Imhoff et al. 2021). Similarly, exposure to a generalized conspiratorial view of society fosters support for autocratic over democratic governance (Papaioannou et al. 2023).

If the *consequences* of conspiracy theories on political outcomes are well documented, how political contexts influence conspiracy beliefs is much less understood. On the basis of a machine learning analysis of cross-cultural data, Douglas et al. (2023) recently argued that conspiracy beliefs may be even more prevalent in well-functioning societies than in the dysfunctional ones. Indeed, endorsing conspiracy theories may not only emerge in effectively governed societies as expressions of concerns of societal decline, but also because these beliefs require the right to express dissenting thoughts (Douglas et al. 2023). According to this view, high levels of freedom of speech may lead to *increased* conspiracy beliefs. In line with this argument, a recent cross-cultural study found that the relationship between anti-immigration conspiracy beliefs and conventional collective action was stronger in democratic societies (Thomas et al. 2025), highlighting the role of well-functioning societies in stimulating conspiracy beliefs and their consequences.

However, there is also evidence that certain factors indicative of weakened democratic societies strengthen conspiracy beliefs. For example, the level of corruption within one's country is a robust predictor on conspiracy beliefs (Alper 2023; Alper et al. 2024; Cordonier and Cafiero 2024). Furthermore, corruption has been shown to narrow the gap in conspiracy mentality typically observed between right- and left-wing individuals (Alper and Imhoff 2023). The authors interpreted this effect as resulting from corruption increasing the plausibility of conspiracy theories (see also Schnaudt (2024), for similar findings on the influence of the quality of elections on conspiracy beliefs). A similar dynamic may be at play in societies with impeded freedom of speech, in which one may reasonably be suspicious that important information is deliberatively concealed by authorities.

Less egalitarian societies, characterized by greater economic inequality, are also more susceptible to conspiracy beliefs (Becchetti et al. 2023; Casara et al. 2022; Hornsey, Pearson, et al. 2023; Jetten et al. 2022). In this context, conspiracy theories have been described as stemming from the negative moral evaluations of economically advantaged citizens (Zeng et al. 2024). Besides, conspiracy beliefs reduce the tendency to internally attribute inequalities among members of disadvantaged groups—making such beliefs psychologically adaptive in societies characterized by important inequalities (Nera et al. 2024).

Therefore, citizens' sense of empowerment and control appears to be crucial in preventing conspiracy beliefs (Dow et al. 2022; Küppers and Reiser 2024; van Prooijen 2018). For example, perceptions of low procedural justice on the part of authorities have been related to increased conspiracy beliefs in Australia during the COVID-19 pandemic (Murphy 2024). Similarly, conspiracy believers favour direct democracy over alternative decision-making processes (Küppers 2024). This tendency was confirmed in an experimental design, which showed that making direct democracy salient in a fictional society (i.e., increasing participants' voice), as opposed to representative democracy, reduced their conspiracy beliefs (Pantazi et al. 2022, Study 3). That is, when individuals felt that they directly took part in political decisions, they were less suspicious of political elites and other typical actors of conspiracy theories. Hence, beyond the perception that individuals have on the legitimacy and impact of the expression of their opinions, the global perception that a society respects and values freedom of speech may be key in *reducing* conspiracy beliefs.

We decided to rely on this evidence to inform our predictions. That is, in line with Pantazi et al. (2022) and contrasting with Douglas et al. (2023), we hypothesized that perceiving that one's country values and protects freedom of speech should reduce people's endorsement of conspiracy beliefs. Conversely, perceiving the state of freedom of speech as declining should increase conspiracy beliefs.

2 | The Current Studies

Across five studies, we investigated the relationship between freedom of speech and conspiracy beliefs at the country-level (Study 1) and using experimental designs (Studies 2a–3b). We also systematically controlled for participants' perception of democracy to isolate the specific effect of freedom of speech from the potential confounding influence of their overall perceptions of democracy.[1] A summary of the designs, sample sizes and key findings is provided in Table 1.

In all studies, unless stated otherwise, participants had to indicate their level of agreement with the statements using a 5-point response scale ranging from 1 = *strongly disagree* to 5 = *strongly agree*. All of the a priori power and sensitivity analyses were conducted on G*Power (Faul et al. 2007; version 3.1.9.4). All participants gave their informed consent before starting the experiment.

3 | Study 1

No published studies have investigated the relationship between freedom of speech and conspiracy beliefs. Hence, we first tested

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TABLE 1 | Summary of the designs, sample sizes and key findings.

	Design	Framing	Sample	Findings
Study 1	Correlational (country-level)	/	$N = 68$ countries	Negative bivariate correlation between freedom of speech and conspiracy beliefs. This relationship was not robust to the inclusion of confounders (i.e., corruption and electoral democracy)
Study 2a	Experimental (preregistered); two conditions: high vs. low freedom of speech	Fictional society	$N = 190$ participants	Higher levels of conspiracy beliefs in the low vs. high freedom of speech condition. This effect was robust to the inclusion of perceived democracy
Study 2b	Experimental (preregistered); two conditions: high vs. low freedom of speech	Fictional society	$N = 90$ participants	Higher levels of conspiracy beliefs in the low vs. high freedom of speech condition. This effect was robust to the inclusion of perceived democracy
Study 3a	Experimental (preregistered); three conditions: high vs. low vs. stable freedom of speech (control)	Real-world present	$N = 436$ participants	No effect of freedom of speech on conspiracy beliefs; potential credibility issues suspected
Study 3b	Experimental (preregistered); three conditions: high vs. low vs. stable freedom of speech (control)	Real-world future projection	$N = 498$ participants	Higher levels of conspiracy beliefs in the low vs. high and control freedom of speech conditions. This effect was robust to the inclusion of perceived democracy

this link at a country, macro-level using secondary data. Importantly, we distinguished support for freedom of speech in the population, from a more objective indicator of country-level freedom of speech created by experts. By doing so, we sought to account for a broader picture of the potential link between freedom of speech and conspiracy beliefs. Indeed, individuals living in a society where freedom of speech is structurally low might be more suspicious of the official account of major events. But it might also be the case that populations with a lower support for freedom of speech are less tolerant of dissident voices and more suspicious of the related outgroups (e.g., political opponent, minorities and foreign powers). This study was exploratory, and no hypotheses were preregistered.

3.1 | Methods

3.1.1 | Sample, Procedure and Measures

We combined several existing datasets to test our hypotheses, which are described below along with each variable. All analyses were conducted at the country level, which served as our unit of analysis. Depending on the dataset, the number of countries included ranged from 26 to 68 (see specific sample sizes below). Sensitivity analyses for a linear multiple regression design with six predictors ($\alpha < 0.05, 1 - \beta = 0.80$) indicated that, depending on the sample size, we had 80% power to detect effects ranging from $f = 0.35$ (support for freedom of speech; 26 countries) to $f = 0.58$ (expert-estimated freedom of speech; 68 countries).

3.1.1.1 | Conspiracy Beliefs. We used data from Azevedo et al. (2023). Collected in 2020 during the COVID-19 pandemic, these data integrate 69 countries (N = 51,404) and include a four-item conspiracy beliefs scale about COVID-19 (e.g., the coronavirus [COVID-19] is a bioweapon engineered by scientists; $\alpha = 92$; see Azevedo et al. (2023) for the detailed internal reliability statistics per countries). We used an aggregated version of the data, relying only on the mean conspiracy belief score for each country.

3.1.1.2 | Freedom of Speech. We used two indexes of freedom of speech: support for freedom of speech (bottom-up evaluation from the populations) and rated freedom of speech (top-down evaluation by external experts), retrieved from distinct dataset.

3.1.1.2.1 | Support for Freedom of Speech. We used Justitia's 'Free speech index' (2021), which is based on data collected by YouGov in February 2021. This scale measures how much individuals value freedom of speech via eight items (e.g., 'How important is it that people can say what they want without government censorship in our country?', $\alpha = 93$; see Skaaning and Krishnarajan (2021) for detailed materials and sample descriptions). The data were collected by YouGov from 33 countries ($N = 49{,}998$) and matched 26 countries from the conspiracy beliefs dataset. Similarly to conspiracy beliefs, we used an aggregated version of the data, relying on the mean score of support for freedom of speech in each country. Note that we did not have

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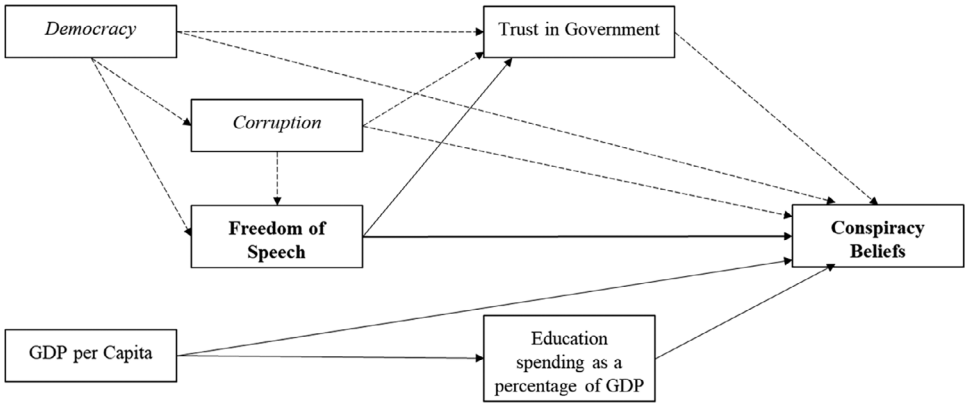

FIGURE 1 | Directed acyclic graph of the hypothesized causal relationships among freedom of speech, conspiracy beliefs and potential covariates. The total effect of interest is indicated by the bold line and variables. Variables in italics represent confounders that should be controlled for to close biasing paths, which are shown as dashed lines.

access to individual items, which impeded calculation of internal consistency.

3.1.1.2.2 | Expert-Estimated Freedom of Speech. We used the 'Freedom of expression index' of the Varieties of Democracy (V-Dem) institute. This index estimates on a year-to-year basis the 'extent to which people can discuss political matters at home and in the public sphere, the press and media are free and can present different political perspectives, and the freedom of academic and cultural expression'. (V-Dem 2022a). The data are an aggregation of expert evaluations, and we used the data from 2020 to match the COVID-19 conspiracy beliefs measure. Overall, 68 countries from the conspiracy beliefs dataset could be matched.

3.1.1.3 | Selection of Control Variables. The relationship between freedom of speech and conspiracy beliefs has not yet been empirically described in the literature. Therefore, we sought to ensure that the predictive power of freedom of speech is distinct from that of related variables whose relationships with conspiracy beliefs have already been documented. We selected a range of candidate variables whose inclusion in the model may be crucial for isolating the total effect of freedom of speech on conspiracy beliefs. Specifically, we considered electoral democracy, trust in government, corruption, GDP per capita and education spending as a percentage of GDP.

To select only the relevant covariates and minimize confounding bias, we generated a directed acyclic graph (DAG) using DAGgity (Textor et al. 2016) to model the causal relationships between our key variables (freedom of speech and conspiracy beliefs) and the covariates. As indicated by the DAG, only two covariates emerged as confounders—corruption and trust in government—and should be controlled for to *close* the related biasing paths (see Figure 1). The two variables are described below.

By contrast, trust in government was conceptualized as a mediator of corruption, electoral democracy and freedom of speech within countries and should not be controlled for to avoid overcontrol bias (Rohrer 2018). Lastly, GDP per capita and education spending as a percentage of GDP may be causally related to conspiracy beliefs, but we have little reason to expect that they are also related to freedom of speech. Therefore, these two variables should not be controlled for either. A description of the covariates not included in the model can be found in the Supporting Information section.

3.1.1.3.1 | Corruption. Increased corruption has been associated with a degradation of press freedom across countries (Hamada et al. 2019; Solis and Antenangeli 2017), which is indicative of lower freedom of speech. We conceptualized corruption as a confounder, as it is also strongly linked to conspiracy beliefs. Indeed, the relationship between the level of corruption at the country-level and conspiracy beliefs has recently received extensive support (Alper 2023; Alper and Imhoff 2023; Cordonier and Cafiero 2024), with conspiracy beliefs being systematically higher in more corrupted countries. In line with previous studies on the topic (e.g., Alper 2023), we controlled for the effect of corruption by relying on the Corruption Perception Index, which ranks countries according to the perceived level of corruption in their public sectors from 0 to 100 (0 indicates extreme corruption; Transparency International 2020). We used the 2020 edition of the index (Transparency International 2020), to match the temporality of our other measures. These data include 68 countries.

3.1.1.3.2 | Electoral Democracy. Perceptions of freedom of speech are deeply rooted in the perceptions of how democratic a society is (Doré et al. 2022). To test whether potential relationships between freedom of speech and conspiracy beliefs are distinct from the mere perceptions of democracy, we included an index of electoral democracy as control variable, also retrieved from the data of the V-Dem institute. This index is estimated by experts and captures whether 'political leaders are elected under comprehensive suffrage in free and fair elections, and freedoms of association and expression are guaranteed' (V-Dem 2022b), on a scale ranging from 0 to 1 (most democratic). All countries from the conspiracy beliefs dataset could be matched.

3.2 | Results

Descriptive statistics and correlation coefficients are presented in Table 2. A similar table, including candidate variables not selected for the model as they were not confounders, can be found in Table S1.

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TABLE 2 | Means, standard deviations and Pearson's correlation coefficients (Study 1).

	M	SD	1.	2.	3.	4.	5.
1. Conspiracy beliefs	3.06	1.18	—				
2. Support for freedom of speech	61.2	13	−0.58**	—			
			($k = 26$)				
3. Expert-estimated freedom of speech	74.9	24.9	−0.29*	0.48**	—		
			($k = 68$)	($k = 33$)			
4. Electoral democracy	63.7	24.1	−0.38**	0.67***	0.92***	—	
			($k = 33$)	($k = 33$)	($k = 78$)		
5. Corruption	50.6	20.1	−0.38**	0.66***	0.46***	0.65***	—
			($k = 68$)	($k = 33$)	($k = 78$)	($k = 78$)	

Note: Below each correlation coefficient is the number of countries that could be matched. For example, the correlation between conspiracy beliefs and support for freedom of speech is based on 26 countries for which data could be retrieved.
*$p < 0.05$.
**$p < 0.01$.
***$p < 0.001$.

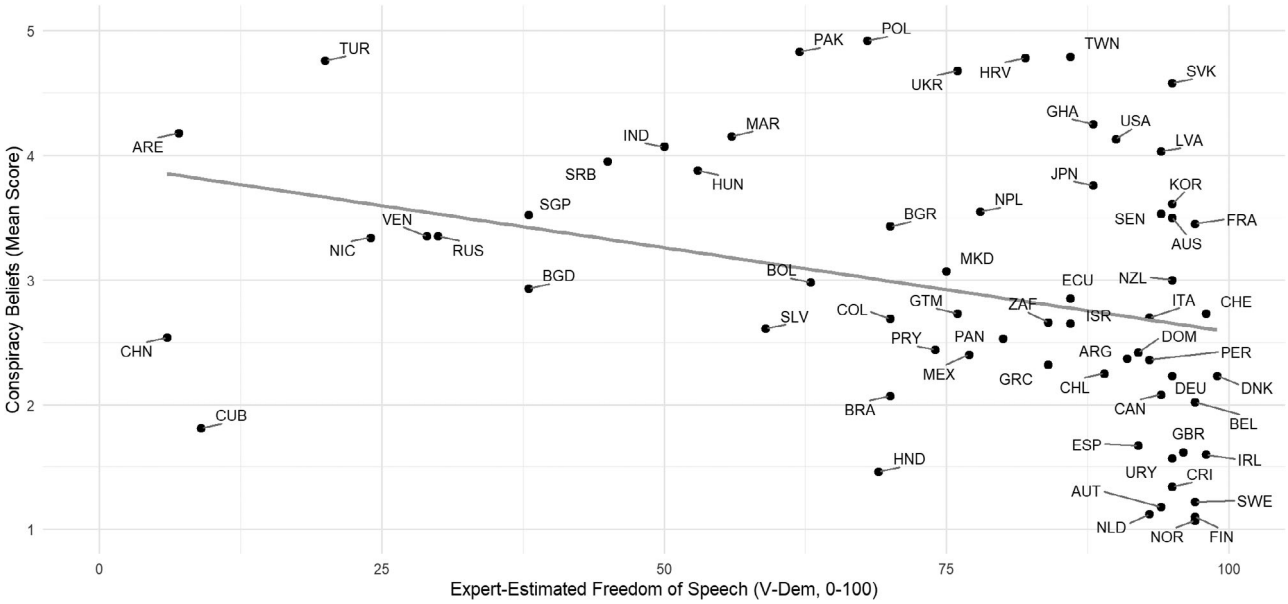

FIGURE 2a | The relationship between expert-estimated freedom of speech and conspiracy beliefs. *Note:* $N = 68$ countries. Countries are indicated by their ISO codes.

Expert-estimated freedom of speech and support for freedom of speech were negatively and significantly associated with COVID-19 conspiracy beliefs (see Figure 2a, 2b).

Next, we conducted hierarchical regressions to further examine the relationship between the two indices of freedom of speech and COVID-19 conspiracy beliefs while controlling for the confounders identified through the DAG. Specifically, in Step 2, we included the first confound identified through the DAG, namely, the country-level level of corruption. Then, in Step 3, we added electoral democracy.

The relationships between freedom of speech indices and conspiracy beliefs were no longer significant when controlling for the confounding variables in Steps 2–3, although the *p* value of support for freedom of speech remained indicative of a potential signal. Statistical details are reported in Table 3. Additional hierarchical regressions, with both indices of freedom of speech entered simultaneously, as well as analyses, including the covariates not selected following the DAG, are reported in Tables S2 and S3, respectively.

3.3 | Discussion

Overall, this first study provided preliminary evidence for the negative link between freedom of speech and conspiracy beliefs. Using secondary analysis of country-level data, our findings partly support the idea that individuals are more prone to endorse conspiracy theories about the COVID-19 in countries where freedom of speech is less developed, and in countries where the population is less supportive of freedom of speech. However, the

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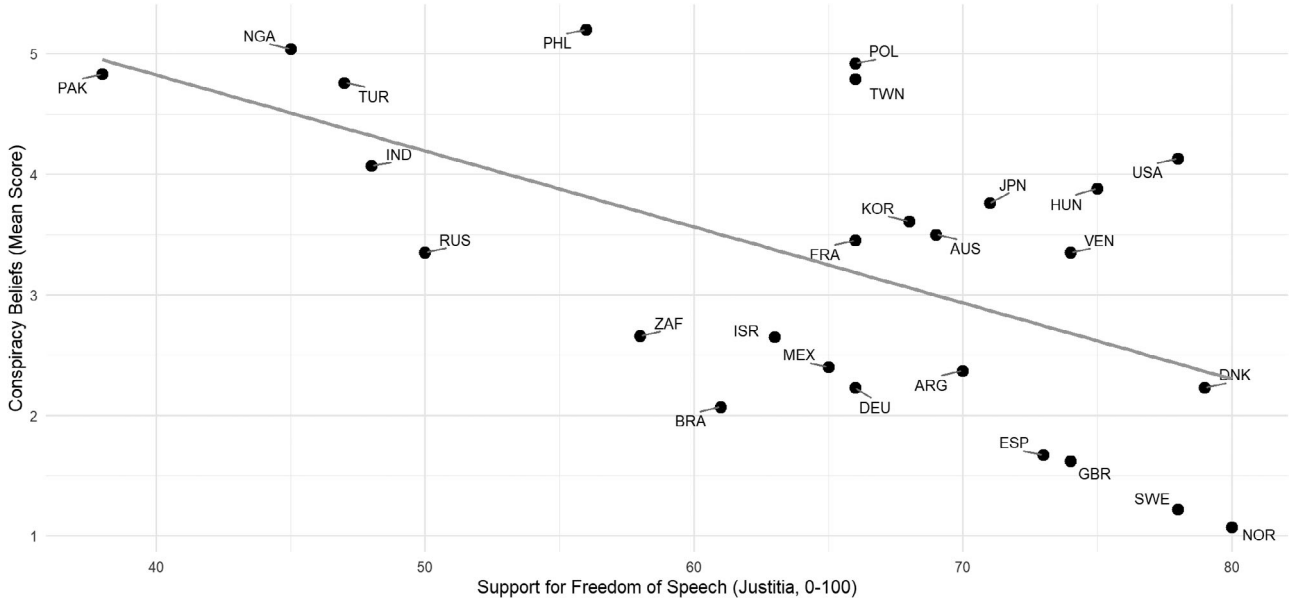

FIGURE 2b | The relationship between support for freedom of speech and conspiracy beliefs. *Note:* $N = 26$ countries. Countries are indicated by their ISO codes.

TABLE 3 | Hierarchical regression of expert-estimated freedom of speech (left) and support for freedom of speech (right) on conspiracy beliefs, controlling for corruption and electoral democracy.

Step 1	***β***	**95% CI**	***t***	***p***	**Step 1**	***β***	**95% CI**	***t***	***p***
Expert-estimated freedom of speech	−0.29	[−0.53; −0.55]	−2.46	0.02	Support for freedom of speech	−0.58	[−0.92; −0.24]	−3.48	0.002
Step 2	*β*	95% CI	*t*	*p*	**Step 2**	*β*	95% CI	*t*	*p*
Expert-estimated freedom of speech	−0.14	[−0.40; 0.11]	−1.12	0.27	Support for freedom of speech	−0.48	[−0.93; 0.03]	−2.17	0.04
Corruption	−0.31	[−0.57; −0.58]	−2.44	0.02	Corruption	−0.16	[−0.61; 0.29]	−0.73	0.47
Step 3	*β*	95% CI	*t*	*p*	**Step 3**	*β*	95% CI	*t*	*p*
Expert-estimated freedom of speech	0.33	[−0.37; 1.03]	0.94	0.35	Support for freedom of speech	−0.40	[−0.87; 0.06]	−1.80	0.09
Corruption	−0.16	[−0.49; 0.18]	−0.95	0.35	Corruption	0.16	[−0.05; 0.81]	0.49	0.63
Electoral democracy	−0.59	[−1.39; 0.22]	−1.45	0.15	Electoral democracy	−0.43	[−1.09; 0.23]	−1.35	0.19

Note: We report standardized coefficients.

relationships were largely attenuated when controlling for two potential confounders—corruption and electoral democracy.

Inferences from this study should be drawn cautiously, especially because the *p* values for freedom of speech indices at Step 1 fell just below the significance threshold. Moreover, we relied on aggregated, country-level measures of conspiracy beliefs and support for freedom of speech, and the study was limited by low statistical power. This lack of statistical power may have created suboptimal conditions for investigating the role of freedom of speech in shaping conspiracy beliefs. The fact that the well-documented indices used as covariates were themselves unrelated to conspiracy beliefs in Step 3 supports this interpretation. In other words, although the evidence from this first study is insufficient to determine whether an effect exists, it would also be premature to conclude that there is no signal worth further investigation.

In the following experimental studies, we focused on freedom of speech as a societal state, rather than as a political conviction (i.e., support for freedom of speech). We tested the hypothesis that perceived low freedom of speech causally increases conspiracy beliefs. Due to the high correlation between expert-estimated freedom of speech and electoral democracy, we systematically controlled for perceptions of democracy in all subsequent studies as part of our exploratory analyses.

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4 | Study 2a

Building upon Study 1, this study tested the hypothesis that inducing low freedom of speech increases conspiracy beliefs. Experimental research poses significant challenges in the study of conspiracy beliefs (Sassenberg et al. 2023). Therefore, we relied on a fictional society paradigm, a method commonly used to investigate such beliefs (e.g., Bertin 2024; Bertin et al. 2024). Specifically, we manipulated the level of freedom of speech (high vs. low) in a fictional country and measured the level of conspiracy beliefs about a specific event occurring in this fictional country. We preregistered the study[2]: https://osf.io/qatk9.

4.1 | Methods

4.1.1 | Participants

We based our sample size on an a priori power analysis conducted via the pwr R-package (Champely 2020). As we could not inform our effect size estimate on previous experimental findings, we opted for a medium effect size (Lovakov and Agadullina 2021), as this strikes a balance between statistical power and resources. Given a two-group between-participants design, an effect size of $d = 0.4$, $\alpha < 0.05$ two-tailed and $1 - \beta = 0.80$, we needed at least $N = 100$ participants per condition. In light of this, we aimed for 230 participants.

We managed to recruit 202 English-speaking American participants on Prolific Academic. In line with preregistration, we excluded 12 participants who failed the attention check, seriousness check, quality check ('When answering the above questions, did you really take the perspective of a citizen of Tribele?'), or whose participation time fell outside of the 3MAD preregistered criterion (Leys et al. 2019). The final sample comprised 190 participants (79 females). Age was assessed through categories: 18–30 ($N = 66$), 30–40 ($N = 62$), 40–50 ($N = 38$), 50–60 ($N = 12$), 60–70 ($N = 9$), above 70 ($N = 3$). The high freedom of speech condition comprised 90 participants, and the low freedom of speech comprised 97 participants.

4.1.2 | Procedure

We randomly assigned participants to either the high or the low freedom of speech condition. In both conditions, participants had to imagine themselves being citizens of a fictional society, Tribele, and to examine a series of articles describing life in the country and narrating a recent event. Participants read a news article describing Tribele's level of freedom of speech.

In the low freedom of speech condition, participants read about a society that keeps its citizens under surveillance, that scrutinizes and filters their opinions online and that persecutes journalists. In the high freedom of speech condition, participants read about how the citizens of Tribele can share their opinions freely, how peaceful protests take place, and how the news media are independent from the government. Participants then answered manipulations checks, indicating the extent to which they believed that freedom of speech prevailed in Tribele and how democratic they believed the country to be.

Next, participants read a second bogus article describing a fatal car crash in which a Tribele politician died. The accidental death of officials is a frequent ground for conspiracy theories (e.g., the Smolensk plane crash; see Cichocka et al. (2016), Study 2; Pantazi et al. 2022, Study 2). In this experiment, the victim was the leader of Tribele's opposition party and the favourite to the upcoming national elections. The article mentioned that although the government had officially declared the event an accident, a fellow party member suspected the government to be involved. Participants were then asked to indicate the degree to which they agreed with this conspiratorial interpretation of the event. After having answered to sociodemographic questions, participants were debriefed and paid for their participation.

4.1.3 | Measures

4.1.3.1 | Perception of Freedom of Speech. As a manipulation check, participants rated to what extent they perceived that freedom of speech existed in Tribele (1 = inexistent; 2 = low; 3 = neither low nor high; 4 = high; 5 = absolute).

4.1.3.2 | Perception of Democracy. As in Study 1, we included a measure of perception of the level of democracy, which is intrinsically linked to freedom of speech. This measure was especially useful to isolate the specific effect of freedom of speech from the more general influence of perceptions of democracy. We measured to what extent participants perceived Tribele to be a democratic society (1 = highly antidemocratic; 2 = mostly antidemocratic; 3 = neither antidemocratic nor democratic; 4 = mostly democratic; 5 = highly democratic).

4.1.3.3 | Event-Specific Conspiracy Beliefs. We used four items adapted from van Prooijen and Jostman (2013) to measure conspiracy beliefs about the car crash (e.g., 'I am inclined to believe that the car was sabotaged'; $\alpha = 0.87$).

4.2 | Results and Discussion

Descriptive statistics and correlation coefficients are displayed in Table 4.

4.2.1 | Manipulation Checks

We tested all our hypotheses using Welch's *t*-tests (Delacre et al. 2017). We first ensured that our experimental manipulation worked. As expected, participants in the high freedom of speech condition perceived a higher level of freedom of speech in Tribele ($M = 3.89$, SD = 1.29) compared to participants in the low freedom of speech condition ($M = 2.05$, SD = 1.29), Welch's $t\ (155) = -12.00$, $p < 0.001$, $d = -1.74$. The effect remained significant after accounting for perceptions of democracy, $F(1, 184) = 8.13$, $p = 0.005$, $\eta^2_p = 0.42$.

The perceived level of democracy was also higher in the high ($M = 4.06$, SD = 0.89) than in the low freedom of speech condition ($M = 2$, SD = 1.36), Welch's $t\ (167) = -12.3$, $p < 0.001$, $d = -1.79$.

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TABLE 4 | Means, standard deviations and Pearson's correlation coefficients (Study 2a).

	M	SD	1.	2.	3.
1. Perception of freedom of speech	2.94	1.41	—		
2. Perception of democracy	3	1.55	0.87***	—	
3. Event-specific conspiracy beliefs	4.75	1.33	−0.27***	−0.35***	—

Note: $N = 190$.[3] All variables were measured using a 7-point Likert scale.
***$p < 0.001$.

4.2.2 | Conspiracy Beliefs

As predicted, participants expressed a higher level of conspiracy beliefs under low freedom of speech (M = 5.37, SD = 1) than under high freedom of speech (M = 4.07, SD = 1.31), Welch's t (166) = 7.57, $p < 0.001$, $d = 1.11$. This effect remained significant after accounting for perceptions of democracy, $F(1, 184) = 28.149$, $p < 0.001$, $\eta^2_p = 0.13$.

In line with our hypotheses, the perceived level of freedom of speech in the society in which a negative event took place affected the endorsement of a conspiratorial view of the event. That is, if one perceives that information in general is filtered by authorities, one will be more inclined to endorse the conspiratorial explanations of this negative event.

Replications are key to ensure the robustness of findings (Nosek et al. 2022). Therefore, in the next study, we tried to replicate our experimental findings. We also switched from a North American to a Belgian population to ensure generalizability of the effect of low freedom of speech on conspiracy beliefs. Last, we extended our measures to conspiracy mentality to further test the scope of the freedom of speech effect.

5 | Study 2b

Study 2b was a direct replication of Study 2a. We also integrated a measure of generic conspiracy beliefs to test whether the pattern of findings of Study 2a generalizes to generic conspiracy beliefs. This study was preregistered: https://osf.io/7ucfz.

5.1 | Methods

5.1.1 | Participants

An a priori power analysis revealed that replicating the effect size found in Study 2a (i.e., d = 1.1) would require at least N = 76 participants (given a two-condition between-subjects design, $1 - \beta = 0.90$, and $\alpha = 0.001$). We planned to recruit 100 participants to anticipate for potential exclusions.

We recruited 100 students on the SONA system of the first author's university (Université libre de Bruxelles, Belgium). We excluded from the analyses three participants who failed the attention check and seven who did not express their consent to participate. Hence, the sample comprised 90 participants (13 males, six others, $M_{age} = 23.9$, $SD_{age} = 4.31$, min = 20, max = 42).

5.1.2 | Procedure

The procedure and materials were the same as in Study 2a, except that it was translated into French to fit our French-speaking Belgian sample.

5.1.3 | Measures

In addition to using the same measure of event-specific conspiracy beliefs as in Study 2a (α = 0.80), Study 2b included a measure of generic conspiracy beliefs. Specifically, we adapted the Conspiracy Mentality Questionnaire (CMQ; Bruder et al. 2013), by asking the participants to answer as a citizen of Tribele. This scale comprised five items (e.g., 'Events which superficially seem to lack a connection are often the result of secret activities'; α = 0.90), and participants rated how probable the statement was on an 11-point scale ranging from 0 = 0% certainly not to 10 = 100% certain.

5.2 | Results and Discussion

Table 5 shows the descriptive statistics and correlation coefficients.

5.2.1 | Manipulation Checks

The experimental manipulation was again successful. Participants perceived a higher level of freedom of speech in the high freedom of speech condition (M = 3.76, SD = 0.79) than in the low freedom of speech condition (M = 1.93, SD = 1.04), Welch's t (80.3) = 9.33, $p < 0.001$, $d = 1.97$. As in Study 2a, this effect remained significant when controlling for perceptions of democracy, $F(1, 87) = 12.3$, $p < 0.001$, $\eta^2_p = 0.12$.

The perceived level of democracy was also higher in the high freedom of speech condition (M = 3.87, SD = 0.89) than in the low freedom of speech condition (M = 2.07, SD = 1.19), Welch's t (79.3) = 8.13, $p < 0.001$, $d = 1.72$.

5.2.2 | Conspiracy Beliefs

As expected, participants reported higher event-specific conspiracy beliefs in the low freedom of speech condition (M = 3.70, SD = 0.58) than in a high level of freedom of speech condition (M = 3.32, SD = 0.77), Welch's t (83.7) = −2.67, $p = 0.009$, $d = -0.56$. This effect replicated even when accounting for perceptions of democracy, $F(1, 87) = 5.13$, $p = 0.026$, $\eta^2_p = 0.056$.

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TABLE 5 | Means, standard deviations and Pearson's correlation coefficients (Study 2b).

	M	SD	1.	2.	3.
1. Perception of freedom of speech	2.87	1.30	—		
2. Perception of democracy	2.99	1.38	0.90***	—	
3. Event-specific conspiracy beliefs	3.51	0.71	−0.21*	−0.15	—
4. Generic conspiracy beliefs (CMQ)	7.97	1.82	−0.10	−0.12	0.70***

Note: $N = 90$. All variables were measured using a 5-point scale, except for conspiracy mentality (11 points).
*$p < 0.05$.
***$p < 0.001$.

Furthermore, participants also reported higher scores of generic conspiracy beliefs when in the low freedom of speech condition ($M = 8.38$, SD = 1.51) than in the high freedom of speech condition ($M = 7.57$, SD = 2.02), Welch's $t(83.2) = -2.16$, $p = 0.034$, $d = -0.45$. However, this effect was no longer significant when controlling for perceptions of democracy, $F(1, 87) = 3.36$, $p = 0.07$, $\eta^2_p = 0.037$.

In this study, we replicated Study 2a's findings on a French-speaking Belgian sample. Furthermore, we found that inducing a low level of freedom of speech did not only increase event-specific conspiracy beliefs but also generic conspiracy beliefs—although the latter effect was no longer significant when controlling for perceptions of freedom of speech and may have been primarily caused by the adaptation of the CMQ to the fictional context.

Across Studies 2a–2b, the effect of perceptions of freedom of speech on specific conspiracy beliefs appears robust across contexts (i.e., English-speaking Americans and French-speaking Belgians) and scales (specific conspiracy beliefs and, to a lesser extent, generic conspiracy beliefs, adapted to the fictional context). However, the variations of the level of freedom of speech could be associated with inductions of democratic versus totalitarian regimes in our experiments, hindering their internal validity. Indeed, the way we described the fictional societies could not rule out the interpretation that the increase of freedom of speech we observed was due to the change of political regime. Furthermore, these experiments relied on fictional scenarios, which reduced their ecological validity. Last, we could not determine whether the effect was driven by an increase or decrease in freedom of speech, because we lacked a control condition. Hence, in Study 3a, we sought to conceptually replicate this effect while addressing these issues.

6 | Study 3a

This study aimed at conceptually replicating Studies 2a and 2b in a real-world context while adding a control condition to gain a finer grained understanding of the effects of freedom of speech on conspiracy beliefs. This study was preregistered: https://osf.io/efahy.

6.1 | Methods

6.1.1 | Participants

An a priori power analysis revealed that, given a three-condition between-participants design, $\alpha < 0.05$ two-tailed, and $1 - \beta = 0.95$, we needed at least 354 participants to detect an effect as small as $f = 0.21$ (i.e., the smallest effect size observed in Study 2b). Anticipating for potential exclusions, we planned for 450 participants.

We recruited 452 on Foule Factory, a French-speaking crowdsourcing platform. After excluding 11 participants who did not complete the experiment and five participants who failed the attention check, 436 participants remained in the analyses (237 women, $M_{age} = 44.8$, $SD_{age} = 12.3$, min = 21, max = 80).

6.1.2 | Procedure

We designed a new experimental manipulation to induce different levels of freedom of speech in a real-world context. Participants read that they would participate in an information processing task aimed at improving an infographic. The content of the infographic was inspired by Reporters Without Borders' World Press Freedom Index.[4] We simplified this ranking so that countries pertained to one of three categories: high freedom of speech (i.e., green), medium freedom of speech (i.e., yellow and orange) or low freedom of speech (i.e., red). On the top corner, a focus on the participant's country (France) highlighted the dynamic of score and ranking over the last year (increased, stable or decreased depending on the condition). Across conditions, we manipulated the colour, score, ranking and the evolution of the level of freedom of speech in France. For example, in the high freedom of speech condition, France was presented in green, ranking in 7th place with a good score, which was presented as an improvement compared to 2022 (see Figure 3).

Next, participants were asked a series of questions to ensure that they processed the content of the infographic and the level of freedom of speech in France. We did not preregister exclusion of participants who failed the manipulation reinforcement questions, because our intention was primarily to encourage them to examine the details of the infographic. In total, 68 participants failed at least one of these questions ($N_{high\ freedom} = 14$; $N_{low\ freedom} = 31$; $N_{control} = 23$).[5] Results did not change when excluding these participants (see Supporting Information section for the detailed analyses).

Participants were then presented with the same event-specific and generic conspiracy beliefs scale used in previous studies adapted to the French context, demographics, and were debriefed and paid.

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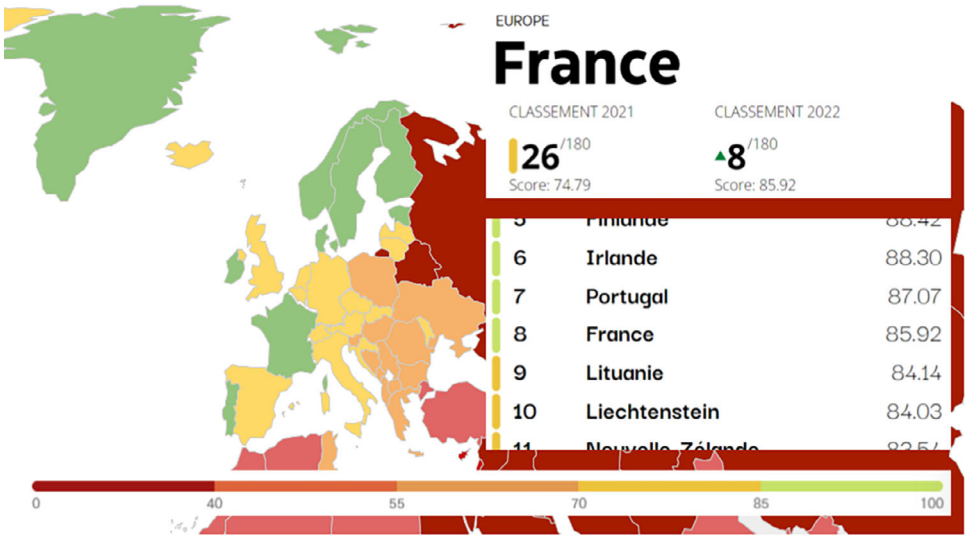

FIGURE 3 | The material used in the high freedom of speech condition.

TABLE 6 | Means, standard deviations and Pearson's correlation coefficients (Study 3a).

	M	SD	1.	2.	3.
1. Perception of freedom of speech	3.55	0.84	—		
2. Perception of democracy	3.69	1	0.70***	—	
3. Event-specific conspiracy beliefs	2.52	0.86	−0.27***	−0.26***	—
4. Generic conspiracy beliefs (CMQ)	7.03	2.17	−0.30***	−0.38***	0.46***

Note: $N = 436$. All variables were measured using a 5-point scale, except for conspiracy mentality (11 points).
***$p < 0.001$.

6.1.3 | Measures

We used the same measures as in Studies 2a and 2b adapted to the French context. The car accident article introducing the specific conspiracy beliefs scale was adapted to refer to a fictional candidate portrayed as the frontrunner in the upcoming internal elections of a rising fictional party. The article was presented as published by a fictional newspaper, but based on information gathered by a real press agency (i.e., Agence France-Presse). Contrary to Study 2b, the CMQ was presented unaltered. The internal reliability of the specific ($\alpha = 0.82$) and generic ($\alpha = 0.90$) conspiracy beliefs scales was good.

6.2 | Results and Discussion

Table 6 shows the descriptive statistics and correlation coefficients.

6.2.1 | Manipulation Check

We tested our hypotheses using the preregistered Helmert contrasts. The first contrast compared the low freedom of speech condition to the two others (−2, 1, 1), and the second contrast compared the high freedom of speech and the control condition (0, −1, 1). The manipulation was successful, as the level of perceived freedom of speech was lower in the low freedom of speech condition ($M = 3.25$, SD = 0.96) compared to the two other conditions ($M = 3.70$, SD = 0.73), $F(1, 433) = 29.80$, $p < 0.001$, $\eta^2_p = 0.064$. The level of perceived freedom of speech did not differ between the high freedom of speech ($M = 3.77$, SD = 0.69) and the control condition ($M = 3.64$, SD = 0.77), $F(1, 433) = 1.75$, $p = 0.186$, $\eta^2_p = 0.004$. When controlling for perceptions of democracy, the effect of low freedom of speech compared to the two other conditions was still significant, $F(1, 432) = 29.29$, $p < 0.001$, $\eta^2_p = 0.06$, and the effect of high freedom of speech compared to the control condition became significant, $F(1, 432) = 6.31$, $p = 0.01$, $\eta^2_p = 0.01$.

The level of perceived democracy decreased in the low freedom of speech condition ($M = 3.54$, SD = 1.02) compared to the two other conditions ($M = 3.77$, SD = 0.99), $F(1, 433) = 5.041$, $p = 0.025$, $\eta^2_p = 0.012$. Again, comparing the high freedom of speech ($M = 3.73$, SD = 0.93) to the control condition ($M = 3.81$, SD = 1.04) yielded no significant difference, $F(1, 433) = 0.45$, $p = 0.50$, $\eta^2_p = 0.001$.

6.2.2 | Conspiracy Beliefs

We applied the same set of contrasts to test our main hypotheses. Against our predictions, we found no significant effect of low freedom of speech ($M = 2.51$, SD = 0.81), compared to other conditions ($M = 2.52$, SD = 0.89), $F(1, 433) = 0.01$, $p = 0.94$, $\eta^2_p = 0.001$.

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Similarly, the level of conspiracy beliefs triggered by the high freedom of speech condition ($M = 2.52$, SD = 0.92) was not significantly different from the control condition ($M = 2.51$, SD = 0.86), $F(1, 433) = 0.01$, $p = 0.91$, $\eta^2_p = 0.001$. The non-significance of these tests remained unchanged after accounting for perceptions of democracy, with the low freedom of speech condition not differing from the two other conditions, $F(1, 432) = 0.46$, $p = 0.50$, $\eta^2_p = 0.01$, and the high freedom of speech not differing from the control condition, $F(1, 432) = 0.01$, $p = 0.94$, $\eta^2_p = 0.01$.

The experimental manipulation had no effect on conspiracy mentality either. Specifically, the low freedom of speech condition ($M = 7.09$, SD = 2.24) did not significantly differ from the two other conditions ($M = 7$, SD = 2.15), $F(1, 433) = 0.15$, $p = 0.70$, $\eta^2_p = 0.001$. The same conclusion applied when comparing the high freedom of speech condition ($M = 7.22$, SD = 2.04) to the control condition ($M = 6.79$, SD = 2.24), $F(1, 433) = 2.88$, $p = 0.09$, $\eta^2_p = 0.007$. This remained the case when controlling for perceptions of democracy, with low freedom of speech condition not differing from the two other conditions, $F(1, 432) = 0.24$, $p = 0.63$, $\eta^2_p = 0.01$, and high freedom of speech condition not differing from the control condition, $F(1, 432) = 2.43$, $p = 0.12$, $\eta^2_p = 01$.

In sum, despite successfully inducing (low) freedom of speech in an ecological setting and observing significant relationships at the cross-sectional level, we did not conceptually replicate the effect of freedom of speech on conspiracy beliefs.[6] These null findings may, of course, be explained by the level of risk associated with testing this new, ecological induction of freedom of speech, as well as by the high probability of obtaining at least one null finding in a series of experiments testing a novel idea (Bertin and Nera 2025; Lakens and Etz 2017).

Another plausible issue is that even though the manipulation of freedom of speech perception was ecological, other aspects of the experimental setting might not have appeared as credible to participants. Notably, participants had no knowledge of the political figure who died in the accident, no knowledge of the political party, and no knowledge of the journal in which this information was published—as all these were made up. In France, such an event would have arguably been discussed in the news.

Therefore, we decided to conceptually replicate Study 3a by adjusting the induction of freedom of speech to increase the credibility of the experiment.

7 | Study 3b

Study 3b was a conceptual replication of Study 3a. The only main difference was the temporal reconfiguration of the experimental material, which was no longer embedded in the present but presented as a future projection task. The study was preregistered: https://osf.io/e9tmx.

7.1 | Methods

7.1.1 | Participants

We kept the same power analysis parameters as the one used for Study 3a (i.e., $f = 0.21$, $\alpha < 0.05$ two-tailed, and $1 - \beta = 0.95$), meaning that we needed at least 354 participants. We decided to boost statistical power by targeting 500 participants, rather than 450, as in Study 3a.

We recruited 500 participants on Prolific. We pre-screened French-speaking participants with a French nationality or living in France to match the samples collected on Foule Factory in previous studies. We excluded two participants who failed the attention check, leaving a total of 498 participants in the analyses (248 women, 5 non-binary, $M_{age} = 34.1$, $SD_{age} = 10.1$, min = 20, max = 75).

7.1.2 | Procedure

We used the exact same procedure as in Study 3a, with the exception that the task was presented as a 'projection in the future' task. Specifically, the induction material was adapted, so that it presented a comparison between the level of freedom of speech in France in 2030 (framed as the present) and the level of freedom of speech in 2024 (framed as the past). Therefore, prior to reading the induction material, participants were asked to imagine themselves in the year 2030. The same instruction was also provided before the measures (i.e., 'Imaginez-vous toujours en 2030'). We also slightly changed the colour of France in the low freedom of speech, so the contrast with the control condition was clearer. Last, we removed one of the reading checks to streamline the procedure.

As in Study 2a, we did not preregister exclusion of participants who failed the manipulation reinforcement questions. In total, 42 participants failed at least one of these questions ($N_{\text{high freedom}} = 12$; $N_{\text{low freedom}} = 11$; $N_{\text{control}} = 19$).[7] Results did not change when excluding these participants (see Supporting Information section for the detailed analyses).

7.1.3 | Measures

We used the same measures as in Study 3a, except that participants were asked to answer to the specific conspiracy beliefs scales while still imagining themselves in the future—in line with the freedom of speech induction. The internal reliability of the specific ($\alpha = 0.83$) and generic ($\alpha = 0.81$) conspiracy beliefs was good. To improve the flow of the survey, the CMQ was no longer measured with an 11-point scale, but rather with a 5-point scale.

7.2 | Results and Discussion

Table 7 shows the descriptive statistics and correlation coefficients.

7.2.1 | Manipulation Check

As in Study 3a, we tested our hypotheses using the preregistered Helmert contrasts. Specifically, the first contrast compared the low freedom of speech condition to the two others (−2, 1, 1), and the second contrast compared the high freedom of speech to the control condition (0, −1, 1). The manipulation was again successful, and the level of perceived freedom of speech was lower

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TABLE 7 | Means, standard deviations and Pearson's correlation coefficients (Study 3b).

	M	SD	1.	2.	3.
1. Perception of freedom of speech	3.57	0.80	—		
2. Perception of democracy	3.66	1.04	0.60***	—	
3. Event-specific conspiracy beliefs	2.82	0.83	−0.24***	−0.26***	—
4. Generic conspiracy beliefs (CMQ)	3.83	0.79	−0.17***	−0.18***	0.47***

Note: $N = 498$. All variables were measured using a 5-point scale.
***$p < 0.001$.

in the low freedom of speech condition (M = 3.24, SD = 0.94) compared to the two other conditions (M = 3.74, SD = 0.67), $F(1, 495) = 44.83$, $p < 0.001$, $\eta^2_p = 0.083$. Moreover, the level of perceived freedom of speech did not significantly differ between the high freedom of speech ($M = 3.77$, SD = 0.67) and the control condition (M = 3.71, SD = 0.68), $F(1, 495) = 0.50$, $p = 0.48$, $\eta^2_p = 0.001$. The effects of the contrasts on the manipulation checks remained unchanged after controlling for perceptions of democracy. Participants in the low freedom of speech still reported more freedom of speech perceptions than in the two other conditions, $F(1, 494) = 23.11$, $p < 0.001$, $\eta^2_p = 0.03$. Also aligned with the preregistered analyses, participants in the high freedom of speech did not report significantly different levels of freedom of speech perceptions than in the control condition, $F(1, 494) = 0.82$, $p = 0.37$, $\eta^2_p = 0.01$.

The level of perceived democracy decreased in the low freedom of speech condition (M = 3.35, SD = 1.10) compared to the two other conditions ($M = 3.81$, SD = 1.10), $F(1, 495) = 21.53$, $p < 0.001$, $\eta^2_p = 0.042$. Comparing the high freedom of speech (M = 3.80, SD = 1.07) to the control condition (M = 3.81, SD = 0.88) again yielded no significant difference, $F(1, 495) = 0.01$, $p = 0.92$, $\eta^2_p = 0.001$.

7.2.2 | Conspiracy Beliefs

In line with our hypotheses, we found a significant effect of low freedom of speech (M = 3.02, SD = 0.81) compared to the other conditions (M = 2.73, SD = 0.82), $F(1, 495) = 14.62$, $p < 0.001$, $\eta^2_p = 0.03$. The second contrast was not significant, as the level of conspiracy beliefs did not differ between the high freedom of speech ($M = 2.75$, SD = 0.81) and the control condition ($M = 2.71$, SD = 0.83), $F(1, 495) = 0.22$, $p = 0.64$, $\eta^2_p = 0.001$. Controlling for perceptions of democracy did not alter the effect of low freedom of speech compared to the two other conditions, $F(1, 494) = 7.51$, $p = 0.006$, $\eta^2_p = 0.02$. Similarly, the effect of high freedom of speech compared to the control condition remained non-significant, $F(1, 494) = 0.22$, $p = 0.64$, $\eta^2_p = 0.01$.

As in Study 3a, the experimental manipulation had no effect on conspiracy mentality. The low freedom of speech condition ($M = 3.46$, SD = 0.77) did not differ from the two other conditions (M = 3.42, SD = 0.79), $F(1, 495) = 0.28$, $p = 0.60$, $\eta^2_p = 0.001$. Regarding the second contrast, the high freedom of speech condition (M = 3.35, SD = 0.82) did not differ from the control condition (M = 3.49, SD = 0.75), $F(1, 495) = 0.11$, $p = 0.74$, $\eta^2_p = 0.005$. This absence of significant findings remained after controlling for perceptions of democracy, with the effect of low freedom of speech (vs. the other two conditions) still not significant, $F(1, 494) = 0.10$, $p = 0.76$, $\eta^2_p = 0.01$. Neither was the effect of high freedom of speech (vs. control condition), $F(1, 494) = 2.64$, $p = 0.11$, $\eta^2_p = 0.01$.

8 | General Discussion

Across five studies, we investigated the relationship between freedom of speech and conspiracy beliefs. In Study 1, we relied on aggregated country-level data from 68 countries and found that lower support for freedom of speech among populations, as well as freedom of speech evaluated by international experts, was associated with increased conspiracy beliefs about COVID-19. However, these relationships were no longer significant after controlling for two confounders—corruption and electoral democracy. In Studies 2a and 2b, we found evidence for a causal impact of the level of freedom of speech on specific and generic conspiracy beliefs. In Study 3a, although a new ecological manipulation of level of freedom of speech proves successful, we did not conceptually replicate this causal effect. In Study 3b, we slightly adjusted our ecological manipulation and confirmed that a decrease in freedom of speech increased conspiracy beliefs. These experimental effects remained unchanged when accounting for perceptions of democracy.

Freedom of speech is a complex construct, as exemplified by the recent four-factor model by Zakharin and Bates (2024). Across this project, we used several conceptualizations of freedom of speech to have a comprehensive view of its effects on conspiracy beliefs. In Study 1, we used both measures of support for freedom of speech and the state of freedom of speech within one's country at a more objective level. Notably, we observed that expert-estimated freedom of speech and electoral democracy were highly correlated.

Therefore, we systematically accounted for perception of democracy in Studies 2a–3b. The experimental effects of freedom of speech observed in Studies 2a–2b and 3b were robust to this control, further supporting the idea that freedom of speech is a specific political factor that fosters conspiracy beliefs.

In Study 3a, freedom of speech was also induced at the country level, but more specifically through freedom of the press. Although this operationalization did not rely on the fictional settings used in Studies 2a and 2b, it sought to increase the internal and ecological validity of our experimental material. Admittedly, the initial operationalization of the vignette and variables may have limited the credibility of our materials, thus

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explaining the observed null findings. We addressed this issue in Study 3b, by keeping the same materials, but operationalizing it as depicting a future, rather than a present state.

This adjustment not only allowed to successfully replicate the effect in an ecological context but also to confirm that this effect was driven by low freedom of speech increasing conspiracy beliefs, rather than high freedom of speech mitigating conspiracy beliefs. This finding also nuances Douglas et al. (2023) claim that well-functioning societies are fertile ground for conspiracy beliefs. In essence, Study 3b's results suggest that ill-functioning societies may be an even more favourable ground for such beliefs. Note also that, although we may consider freedom of speech as an indicator of a society's 'good functioning', what it means for a society to function well is open to debate.

As a set, the present studies further document the macro-level causes of conspiracy beliefs (Hornsey et al., 2018, Hornsey, Bierwiaczonek, et al. 2023) by emphasizing the importance of accounting for freedom of speech. Our findings also extend previous efforts on the cultural underpinning of conspiracy beliefs (Adam-Troian et al. 2021; Douglas et al. 2023) by suggesting that conspiracy theories seem to be more widespread in cultures with lower tolerance for divergent opinions, minority views and media protections. Obviously, the effect of freedom of speech on conspiracy beliefs is compatible with a variety of causal models (e.g., the actual presence of state-sponsored conspiracies may both influence conspiracy beliefs and limit freedom of speech). Future research may want to examine the processes underlying this effect.

Several factors have been proposed to counter the spread of conspiracy theories, ranging from specific interventions (O'Mahony et al. 2023) to broader goals such as improving well-being (Cichocka 2020) and empowerment (van Prooijen 2018). Although we could not provide evidence that enhancing freedom of speech mitigates conspiracy beliefs, ensuring that the state of freedom of speech is not perceived as declining may be one way to anticipate a potential rise in such beliefs. Therefore, boosting the voice given to citizens, the transparency of institutional decisions and the diversity of media coverage are various strategies that may minimize the spread of conspiracy beliefs.

Our studies also come with some limitations. First, we only used aggregated country-level data in Study 1 and only had low statistical power to test our predictions. A second limitation relies on our framing of freedom of speech. Indeed, although we introduced our research as studying the impact of *perceptions* of freedom of speech, the expert-estimated index of Study 1 and the overall framing of the vignettes of Studies 2a–3b rather relied on a contextual, objective approach to freedom of speech. Even though we could argue that we studied the impact of these vignettes on participants' perceptions, future research could clarify the effect of objective freedom of speech situations on perceptions of freedom of speech and their respective impacts on conspiracy beliefs. Third, we relied on WEIRD samples of participants (Henrich et al. 2010). Surveying participants living in contexts with severe censorship seems crucial if one wishes to further document how these kinds of political contexts shape perceptions of social events. Fourth, our analyses controlling for perceptions of democracy were all exploratory and require confirmatory replications. Lastly, the exact processes underlying the effect of freedom of speech on conspiracy beliefs have yet to be documented. Our attempt to account for trust in media as a mediator in Study 3a (see endnote 6) was not fruitful. However, it is unclear whether this was due to issues with the specific conspiracy beliefs measure or the absence of a true effect. In addition to the variables tested in Study 1, such as corruption and trust in government, other factors like perceived lack of political control and feelings of powerlessness may help further explain the effects documented throughout this article. As such, this goal constitutes a promising avenue for future research.

9 | Conclusion

Across five studies, we provided country-level and experimental evidence for the effect of freedom of speech on conspiracy beliefs. We showed that conspiracy beliefs about the COVID-19 were higher in countries with lower levels of expert-estimated and support for freedom of speech. However, the significance of these effects faded after accounting for two confounders, namely, corruption and electoral democracy. Switching to experimental designs, we observed that freedom of speech causally impacted conspiracy beliefs in two studies conducted in a fictional context. This effect was robust to the control of perception of democracy. An initial attempt at replicating this effect with a new ecological manipulation of freedom of speech initially failed, but a refined version of the materials proved successful in a fourth experiment. This last experiment informed us that the effect was driven by low freedom of speech on increasing conspiracy beliefs—rather than high freedom of speech decreasing conspiracy beliefs—and that the effect was again independent from perceptions of democracy. Overall, the present research provides new evidence supporting the importance of open and transparent political environments in fighting the proliferation of conspiracy beliefs.

Author Contributions

Paul Bertin: conceptualized the project, created the materials, conducted the data collections, conducted the analysis, wrote the manuscript and did the project administration and supervision. **Elisa Tognon**: created the materials and wrote the manuscript. **Kenzo Nera**: created the materials, conducted the analysis and wrote the manuscript. **Rrita Bajraktari**: wrote the manuscript. **Pit Klein**: conceptualized the project, created the materials and conducted the data collections. **Vincent Yzerbyt**: wrote the manuscript. **Olivier Klein**: wrote the manuscript.

Acknowledgements

Open access publishing facilitated by Universite de Lausanne, as part of the Wiley - Universite de Lausanne agreement via the Consortium Of Swiss Academic Libraries.

Funding

This work was supported by a grant (PDR 0253.19) from the Belgian National Fund for Scientific Research (FRS-FNRS).

Ethics Statement

This research project has received approval from the ethics committee of the Université libre de Bruxelles (ID: 119/2020) and is in line with the 1964 Helsinki Declaration and its later amendments (2001), the ethical

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principles of the French Code of Ethics for Psychologists (2012) and the American Psychological Association Ethical Principles of Psychologists and Code of Conduct (2017). Participants were informed about the purpose of the studies in a cover letter and were assured that their data would remain confidential. Participants had to give explicit written consent to access the study.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in the Open Science Framework at https://osf.io/nt725/.

Endnotes

[1] We thank the anonymous reviewer for this insightful suggestion. These analyses were exploratory and not preregistered, as they were conducted after all studies had been completed. Therefore, results should be interpreted with caution until confirmatory replications are performed.

[2] The preregistration includes mentions of covariates that were not included in the analyses. We did so to simplify as much as possible the analyses. The results did not differ when including these covariates (see the Supporting Information section).

[3] Three participants did not complete the items measuring perceptions of freedom of speech and democracy. However, we decided to keep these participants in the analyses as they completed the conspiracy beliefs items and passed the various checks. Results did not differ when excluding these participants.

[4] https://rsf.org/en/index.

[5] One of the questions asked participants from the control condition whether freedom of speech had increased, decreased, or stayed the same in France between 2021 and 2022. The correct answer to this question was ambiguous, as we both used an equal sign near the 2022 ranking to indicate no change and slightly increased the score to make it more ecologically valid. Therefore, we excluded only participants who answered that freedom of speech had decreased.

[6] We also preregistered to test whether trust in the media would mediate the effect of freedom of speech, and that we would test this model using continuous scores in the absence of experimental effects. These analyses can be found in the Supporting Information section.

[7] As in Study 3a, we only excluded participants from the control condition who answered that freedom of speech had decreased (see endnote 5).

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